



2022

PRELIMINARY ASSESSMENT TASK 1

Extension 1 Mathematics

General Instructions

- Reading time – 10 minutes
- Working time – 90 minutes
- Write using blue or black pen
Black pen is preferred
- For questions in Section II, show relevant mathematical reasoning and/or calculations.
- Calculators approved by NESA may be used
- DO NOT USE LIQUID PAPER OR CORRECTION TAPE

Total Marks – 60

Section I: 10 marks

- Attempt questions 1-10, using the answer sheet provided.
- Allow about 15 minutes for this section

Section II: 50 marks

- Attempt questions 11-14, answer the questions in the booklets provided.
- Allow about 75 minutes for this section

	M/C	11	12	13	14	TOTAL
Inequalities	2, 3	b (3), e (3)	b (3)	d (3)	e (4)	18
Inverse Functions	5, 10			a (4), b (2), c (2), e (2)	a (8)	20
Introduction to Functions	1, 4, 6					3
Linear, Quadratic and Cubic Functions	7	a (2), c (2)	c (3), d (3)			11
Further functions and relations	8, 9	d (3)	a (3)			8
TOTAL	10	13	12	13	12	60

STUDENT NUMBER: _____ TEACHER: _____

Section I

10 marks

Attempt Questions 1 to 10

Use the multiple-choice answer sheet for Questions 1 to 10

1. Which of the following represents the domain of $f(x) = \sqrt{6 - 2x}$?

(A) $(-\infty, 6]$

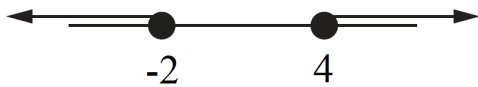
(B) $[3, \infty)$

(C) $(3, \infty)$

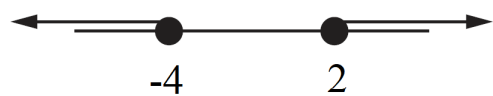
(D) $(-\infty, 3]$

2. Which diagram best represents the solution set of $x^2 - 2x - 8 \leq 0$?

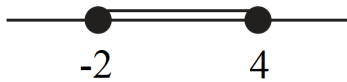
(A)



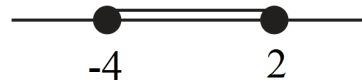
(B)



(C)



(D)



3. Which set provides a solution to the inequality $|3x - 2| \leq |x + 4|$?

(A) $\left[-\frac{1}{2}, 3\right]$

(B) $\left(-\infty, -\frac{1}{2}\right] \cup [3, \infty)$

(C) $\left(-\infty, -\frac{1}{2}\right]$

(D) $(-\infty, 3]$

4. If $f(x) = 2x - 3$, then $f^{-1}(x) =$

(A) $\frac{1}{2}x + 3$

(B) $\frac{1}{2}x + \frac{3}{2}$

(C) $\frac{1}{2x - 3}$

(D) $\frac{1}{2}x - 3$

5. Which of the following functions does not have an inverse function?

(A) $f(x) = (x - 2)^2, D: [0, \infty)$

(B) $f(x) = x^3, D: (-\infty, \infty)$

(C) $f(x) = \sqrt{9 - x}, D: [-3, 3]$

(D) $f(x) = 3x + 7, D: (-\infty, \infty)$

6. The linear function $f(x) = 5 - x$ has range $[-2, 3)$. Which of the following represents its domain?

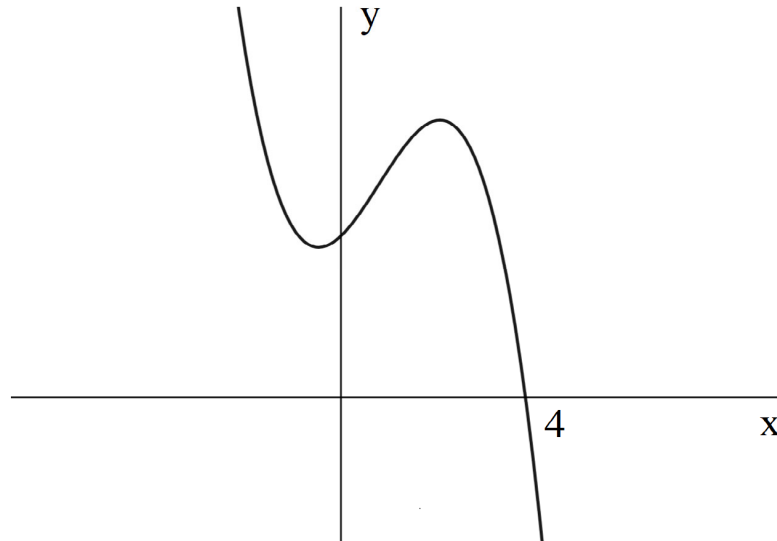
(A) $[-7, 2)$

(B) $(2, 7]$

(C) $[-2, 7)$

(D) $[2, 7)$

7. The graph of the polynomial function is shown below.



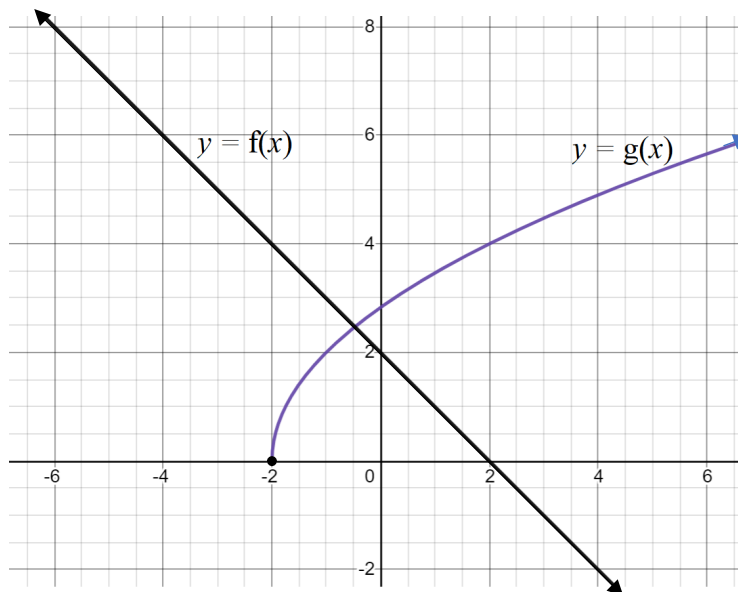
Which of the following could represent the graph above?

- (A) $y = (x - 4)(x^2 + x + 1)$
- (B) $y = (x + 4)(x^2 + x + 1)$
- (C) $y = (4 - x)(x^2 + x + 1)$
- (D) $y = (x - 4)^3$

8. Given the functions $f(x) = 4x + 10$ and $g(x) = \sqrt{x + 2}$, what is the value of a for which $f(g(a)) = 16$?

- (A) $\frac{161}{4}$
- (B) $-\frac{1}{2}$
- (C) $\frac{1}{4}$
- (D) $\frac{9}{2}$

9. The graph below shows the functions for $y = f(x)$ and $y = g(x)$. What are the domain and range of $y = f(x) - g(x)$?



- (A) Domain: $[-2, \infty)$
Range: $(-\infty, \infty)$
- (B) Domain: $(-\infty, \infty)$
Range: $(-\infty, -2]$
- (C) Domain: $(-\infty, -2]$
Range: $(-\infty, 4]$
- (D) Domain: $[-2, \infty)$
Range: $(-\infty, 4]$
10. Given $f(x) = 1 + \frac{2}{x-3}$. Which of the following gives the equations of the horizontal and vertical asymptotes of $f^{-1}(x)$?
- (A) Vertical asymptote is $x = 1$ and horizontal asymptote is $y = 2$
- (B) Vertical asymptote is $x = 1$ and horizontal asymptote is $y = 3$
- (C) Vertical asymptote is $x = 3$ and horizontal asymptote is $y = 1$
- (D) Vertical asymptote is $x = 3$ and horizontal asymptote is $y = 2$

End of Section I

Section II

50 marks

Attempt Questions 11 to 14

Answer each question in a SEPARATE writing booklet.

Extra writing booklets are available.

All necessary working should be shown in every question.

Question 11 (13 marks) Use a SEPARATE writing booklet.

- (a) The coordinates of the turning point of a parabola are $(2, 6)$ and the parabola passes through the point $(3, 3)$. Find the rule for this parabola. **2**

- (b) Solve for x . **3**

$$\frac{4x - 3}{2} - \frac{3x - 3}{3} < 3$$

- (c) For what values of k does $x^2 + (k + 1)x + (k - 2) = 0$ have real roots? **2**

- (d) Solve the following absolute value equation: **3**

$$2|x + 3| = x - 2$$

- (e) Sketch the graphs $y = x^2 - 4$ and $y = \frac{1}{2}|x| - 1$ on the same number plane and write down the inequality that describes the area bounded by the two curves. **3**

End of question 11

Question 12 (12 marks) Use a SEPARATE writing booklet.

- (a) Describe accurately the relation below. **3**

$$x^2 - 8x + y^2 - 14y + 1 = 0$$

- (b) Solve the inequality and draw the solutions on a number line. **3**

$$\frac{6}{5x - 2} < 2$$

- (c) A group of students on a hiking trip decide to travel x km by bus at an average speed of 48 km/h to an unknown destination. They then plan to walk back along the same route at an average speed of 4.8 km/h and to arrive back 24 hours after setting out in the bus. If they allow for a total of 2 hours for lunch and rest, how far must the bus take them? **3**

- (d) Sketch $y = 4x^3 + 3x^2 - 16x - 12$ by first factorising. Label all important features. **3**

End of question 12

Question 13 (13 marks) Use a SEPARATE writing booklet.

- (a) i. Find the inverse of the function with the rule $f(x) = \sqrt{x-2} + 4$. **2**
- ii. Sketch both functions on the same number plane and hence find the point of intersection/s. **2**
- (b) Consider the function $f(x) = \frac{4}{x+2}$.
- Show that $f^{-1}(x) = \frac{4-2x}{x}$. **2**
- (c) Given $f(x) = x^2 - 9$ is a function defined for $x \leq 0$, verify that $f^{-1}(f(x)) = x$. **2**
- (d) Graph the region $x^2 + y^2 \leq 25$, for $x \leq 4$ and $y > -3$. State the domain and range of the region bounded by the inequalities. **3**
- (e) What is the largest positive domain, so that $y = x^2 - 12x + 40$ has an inverse function? **2**

End of question 13

Question 14 (12 marks) Use a SEPARATE writing booklet.

- (a) Consider the function $f(x) = \frac{1}{4}[(x - 1)^2 + 7]$
- i. Sketch the parabola $y = f(x)$. Clearly showing any intercepts with the axes and the coordinates of its vertex in the domain $[-6, 6]$ **2**
 - ii. Find the inverse function, $y = f^{-1}(x)$ with a range of $[1, \infty)$. **2**
 - iii. Find the coordinates of any points of intersection of the two curves $y = f(x)$ and $y = f^{-1}(x)$. **1**
 - iv. Sketch the graph of $y = f^{-1}(x)$ on the same set of axes as your graph in part i. Label the two graphs clearly. **1**
 - v. Let a be a real number not in the domain of $f^{-1}(x)$. Find $f^{-1}(f(a))$. **2**
- (b) For what values of x is the inequality $\frac{1}{x+1} \leq \frac{x+1}{x+3}$ satisfied? **4**

End of paper

MC : Q1 D Q2C Q3A Q4B Q5 A
Q6B Q7C Q8C Q9D Q10 B

Question 11

a)

$$\left. \begin{aligned} y &= a(x-h)^2 + k \\ y &= a(x-2)^2 + 6 \end{aligned} \right\} \textcircled{1}$$

Sub (3,3) into equation

$$3 = a(3-2)^2 + 6$$

$$3 = a + 6$$

$$\therefore a = -3$$

$$y = -3(x-2)^2 + 6$$

$$\text{or } y = -3x^2 + 12x - 6$$

$$b) \frac{4x-3}{2} - \frac{3x-3}{3} < 3$$

$$3(4x-3) - 2(3x-3) < 18$$

$$12x - 9 - 6x + 6 < 18$$

$$6x - 3 < 18$$

$$6x < 21$$

$$x < \frac{21}{6}$$

$$x < \frac{7}{2}$$

$$\frac{3(4x-3) - 2(3x-3)}{6} < 3$$

$$\frac{12x - 9 - 6x + 6}{6} < 3$$

$$6x - 3 < 18$$

$$6x < 21$$

$$x < \frac{7}{2}$$

⑤ if no errors

② if one error

① if two errors

c) $\Delta \geq 0$ for real roots.

$$(k+1)^2 - 4(1)(k-2) \geq 0 \textcircled{1}$$

$$(k^2 + 2k + 1) - 4k + 8 \geq 0$$

$$k^2 - 2k + 9 \geq 0$$

$k^2 - 2k + 9$ is a positive definite parabola

$\therefore \Delta \geq 0$ for all values of k .

$\therefore$ All real values of k .

Alternative:

$$k^2 - 2k + 9 \geq 0$$

$$k^2 - 2k + 1 \geq -9 + 1$$

$$(k-1)^2 \geq -8$$

$\therefore$ True for all real values of k

Since $(k-1)^2 \geq 0$.

Done well.

However, some students used the standard form of quadratic function, $y = ax^2 + bx + c$, and solved the equations simultaneously.

This took more time than the factorised form, $y = a(x-h)^2 + k$.

* Done well

* Some students made algebraic errors but they were given CFE marks depending on the number of mistakes made.

* Large number of students did not simplify their answer.

* Poorly done.

* $\Delta > 0$ was one of the common mistakes.

* Many students managed to build the inequality for $\Delta \geq 0$ but failed to solve it for k .

$$11d) 2|x+3| = x-2$$

$$2(x+3) = x-2 \text{ for } x+3 \geq 0 \quad \text{or} \quad -2(x+3) = x-2 \text{ for } x+3 < 0$$

$$2x+6 = x-2 \quad -2x-6 = x-2$$

$$x = -8 \text{ for } x \geq -3 \quad -3x = 4$$

$$\textcircled{1} \quad x = -\frac{4}{3} \text{ for } x < -3 \quad \textcircled{1}$$

* Most students worked out the two x values of -8 and $-\frac{4}{3}$, however, did not check the solution.

Test solutions

$$2|-8+3| = -8-2$$

$$2|-5| = -10$$

$$10 \neq -10$$

$\therefore x = -8$, not a solution

$$2|-\frac{4}{3}+3| = -\frac{4}{3}-2$$

$$2(\frac{5}{3}) = -\frac{10}{3}$$

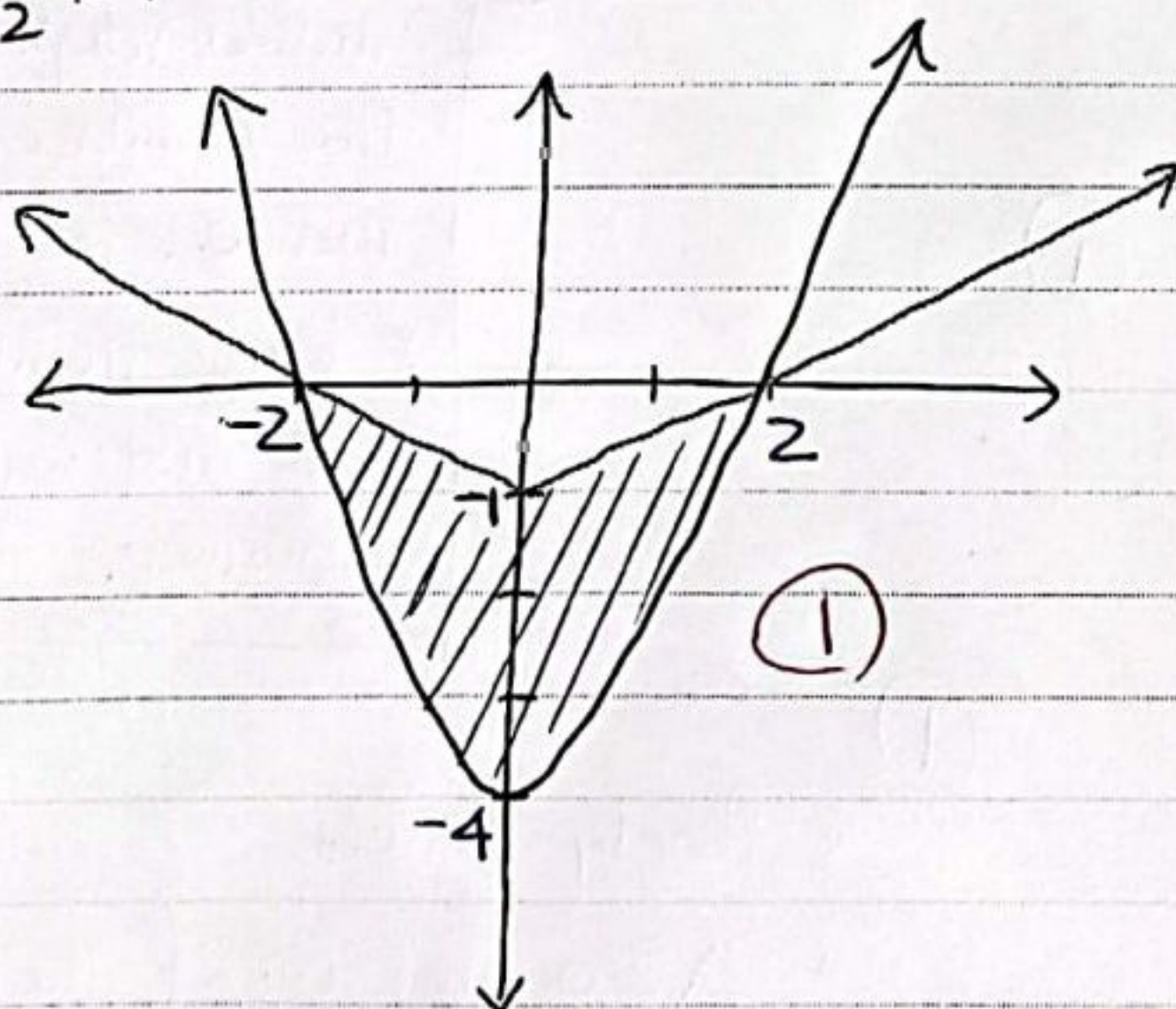
$$\frac{10}{3} \neq -\frac{10}{3}$$

$\therefore x = -\frac{4}{3}$, not a solution $\textcircled{1}$

$\therefore$ no solution

$$e) y = x^2 - 4 \quad [1]$$

$$y = \frac{1}{2}|x| - 1 \quad [2]$$



Substitute $(0, -2)$ into [1],
 $\therefore y \geq x^2 - 4$

Substitute $(0, -2)$ into [2],
 $\therefore y \leq \frac{1}{2}|x| - 1$

$\therefore y \geq x^2 - 4$ and $y \leq \frac{1}{2}|x| - 1$
 $\textcircled{1} \quad \textcircled{1}$

* Poorly done.

* Most students sketched the graphs correctly but could not work out the correct inequalities.

* Many students did not write "and" between the two inequalities.

The area bounded by the curves is the intersection of the two regions

* Common mistake,
 $x^2 - 4 \leq \frac{1}{2}|x| - 1$

This does not represent a region.

$$(12) a) x^2 - 8x + y^2 - 14y + 1 = 0$$

$$x^2 - 8x + \left(\frac{8}{2}\right)^2 + y^2 - 14y + \left(\frac{14}{2}\right)^2 = -1 + \left(\frac{8}{2}\right)^2 + \left(\frac{14}{2}\right)^2$$

$$(x-4)^2 + (y-7)^2 = 64$$

∴ The relation is a circle,
centre (4, 7), radius 8.

It is also a many-to-many
relation.

NOTE: - The question is to describe, it would be a waste of exam time to sketch the graph.

- It is worth 3 marks. Simply describing it as a circle would never be enough for 3 marks. Many students completed the square but didn't use this to help in their description.

$$b) \frac{6}{5x-2} < 2 \quad x \neq \frac{2}{5}$$

$$\times (5x-2)^2 \quad \times (5x-2)^2$$

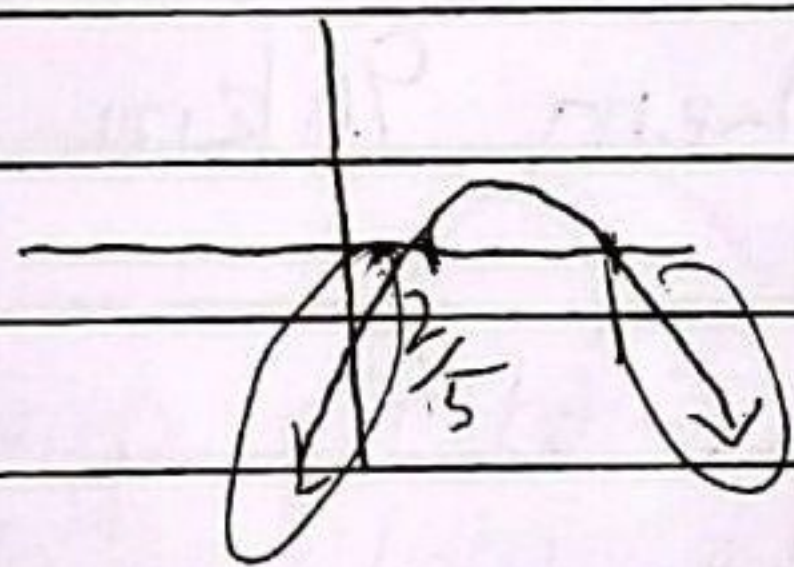
$$6(5x-2) < 2(5x-2)^2$$

$$6(5x-2) - 2(5x-2)^2 < 0$$

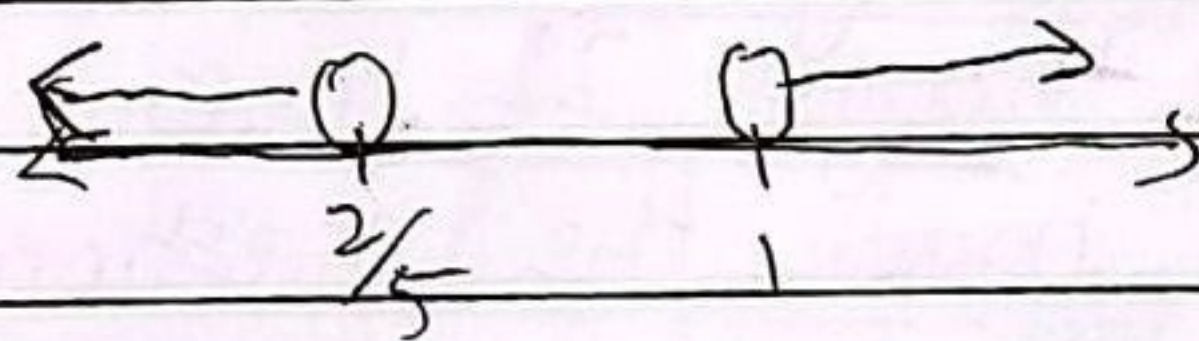
$$(5x-2)(6-2(5x-2)) < 0$$

$$(5x-2)(-10x+10) < 0$$

$$-10(5x-2)(x-1) < 0$$



$$x < \frac{2}{5}, \quad x > 1$$



NOTE: Many students forgot to write the final inequality answer and/or draw it on a number line.

- Lots of students simply multiplied by the denominator instead of its square. This made the question into a linear equation, which meant a max of 2 marks could be given. It also means you have not listened in class and haven't learnt your extension concepts.

- Lots of students changed the LHS and RHS without flipping the inequality symbol. This is incorrect.

$$\begin{aligned} \text{Eg. } 7x+1 &> x^2 \\ x^2 &> 7x+1 \quad \times \\ x^2 - 7x - 1 &> 0 \end{aligned}$$

c) 24 hours total - 2 hours rest
= 22 hours travel time.

$$s = \frac{d}{t} \quad \therefore t = \frac{d}{s}$$

time on bus + time walking = 22 hours

$$\frac{x}{48} + \frac{x}{4.8} = 22$$

$$x + 10x = 1056$$

$$11x = 1056$$

$$x = 96$$

$\therefore$ The bus took them 96 km.

NOTE: Many students misinterpreted the question. Most thought 22 hours was the walking time. This made the question extremely easy ($22 \times 4.8 = 105.6 \text{ km}$). So zero marks were awarded.

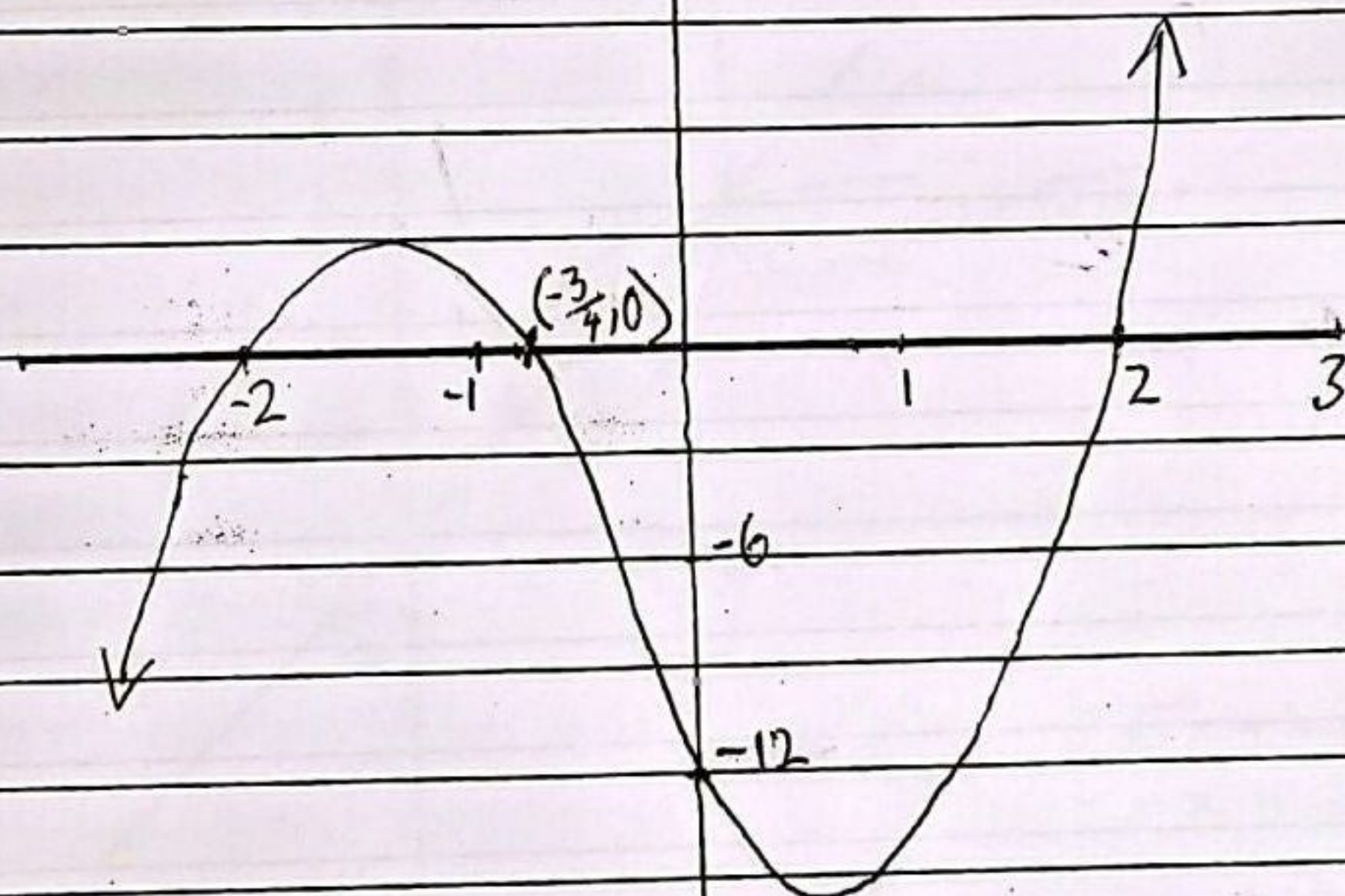
- This question could also have been completed using ratios. 22 hours in the ratio 1:10
 $\therefore$ 2 hours on the bus. (bus: walk)
 $\therefore 2 \times 48 = 96 \text{ km}$.



← Tick this box if you have continued this answer in another writing booklet.

$$\begin{aligned}
 d) \quad y &= 4x^3 + 3x^2 - 16x - 12 \\
 &= x^2(4x+3) - 4(4x+3) \\
 &= (4x+3)(x^2-4) \\
 &= (4x+3)(x-2)(x+2)
 \end{aligned}$$

$$\therefore x\text{-ints} \rightarrow x = -\frac{3}{4}, 2, -2 \quad y\text{-int} \rightarrow y = -12$$



NOTE: The question is to sketch. Therefore it should be your focus. DO NOT draw a tiny graph that is rushed. The graph is the important bit. It should be one third of a page big or more.

- Use a ruler to draw an accurate scale!
- $(x^2-4) \neq (x-4)^2$ a common error.

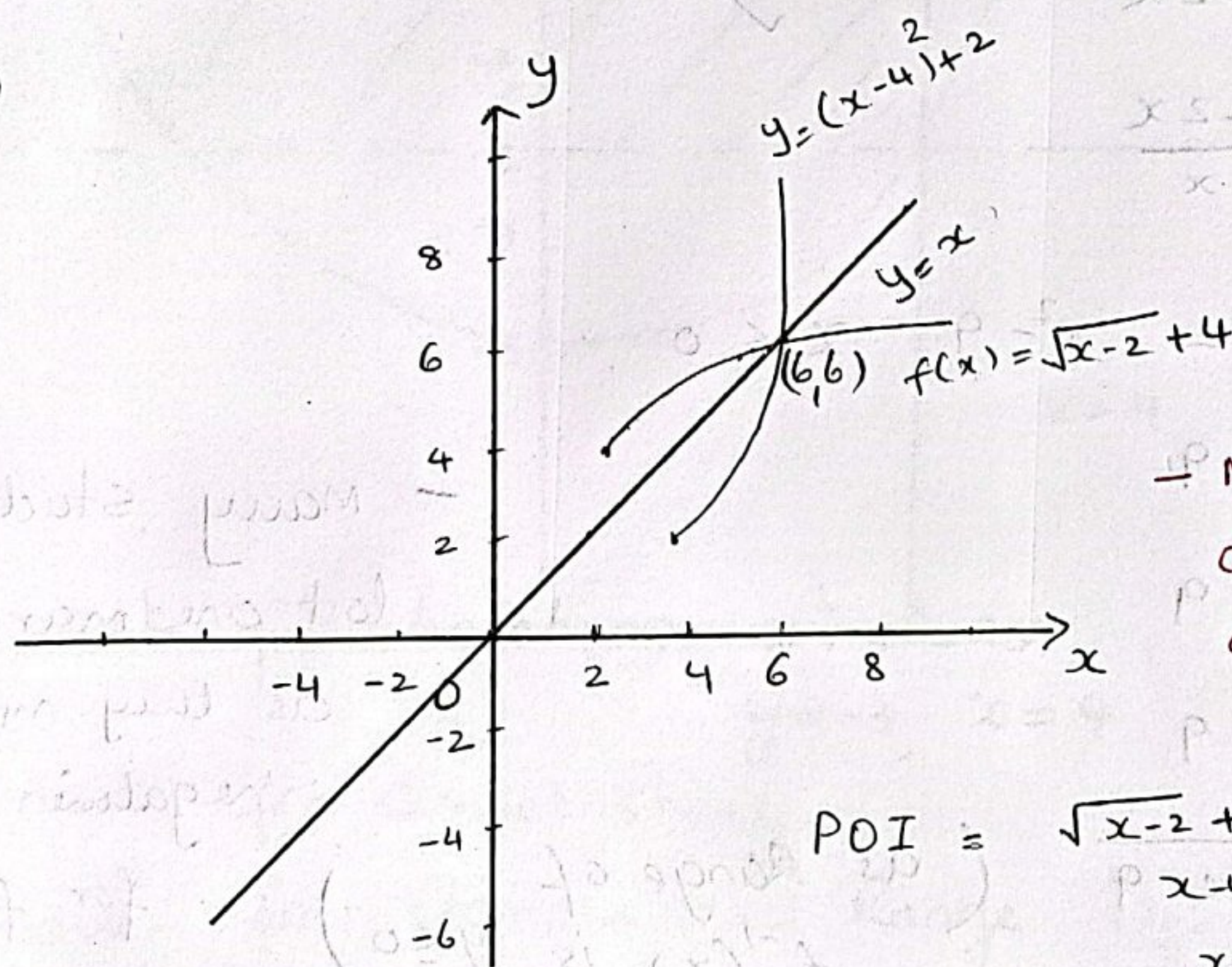
Additional writing space on back page.

- The y-int is not the vertex. The vertex is generally around the middle of the x-intercepts.

Question 13)

- a) i) inverse of $f(x) = \sqrt{x-2} + 4$
- $$y = \sqrt{x-2} + 4$$
- $$x = \sqrt{y-2} + 4 \quad (\text{swap } x \text{ with } y)$$
- $$x-4 = \sqrt{y-2}$$
- $$y-2 = (x-4)^2$$
- $$y = (x-4)^2 + 2$$
- some algebra errors instead of adding 2 at the last step, subtracted 2.

ii)



many students drew both branches of $y = (x-4)^2 + 2$.

$$POI = \sqrt{x-2} + 4 = x$$

$$x-4 = \sqrt{x-2}$$

$$x-2 = x^2 - 8x + 16$$

$$x^2 - 9x + 18 = 0$$

$$(x-6)(x-3) = 0$$

$$(6, 6), (3, 3)$$

(6, 6) is the only Point of Intersection

overall poor attempt.

students are encouraged to attempt more questions like this to improve.

$$b) f(x) = \frac{4}{x+2}$$

$$y = \frac{4}{x+2}$$

$$x = \frac{4}{y+2} \quad (\text{swap } x \text{ with } y)$$

$$xy + 2x = 4$$

$$xy = 4 - 2x$$

$$y = \frac{4-2x}{x}$$

overall,

well done.

$$c) f(x) = x^2 - 9, \quad x \leq 0$$

$$y = x^2 - 9$$

$$x = y^2 - 9$$

$$y^2 = x + 9$$

$$y = -\sqrt{x+9} \quad (\text{as range of } f^{-1}(x) \text{ is } y \leq 0)$$

- many students
lost one mark

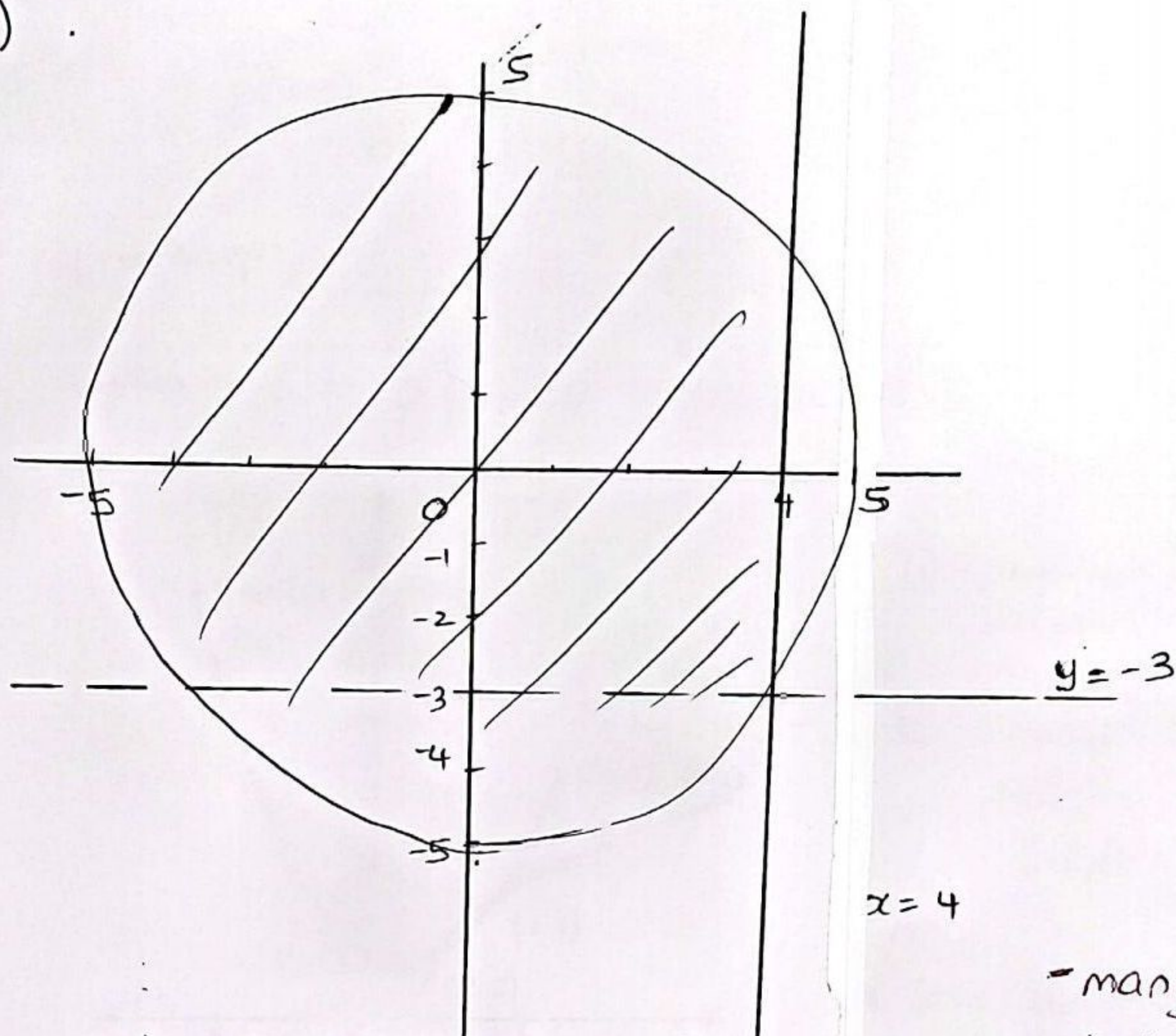
as they missed
negative sign
for $f^{-1}(x)$.

$$\therefore f^{-1}(f(x)) = -\sqrt{x^2 - 9 + 9}$$

$$= -x$$

$$= x \quad \text{as } x \leq 0$$

d) .



1 mark for sketching $x^2 + y^2 = 25$
 $y = -3, x = 4$

1 mark for correct shading

1 mark for domain and Range $D_f: [-5, 4]$

$$R_f = (-3, 5]$$

- many students
lost mark
due to incorrect
shading.

e)

$$y = x^2 - 12x + 40$$

$$\begin{aligned} \text{axis of symmetry} &= -\frac{b}{2a} \\ &= -\frac{(-12)}{2} \\ &= 6 \end{aligned} \quad \checkmark$$

- many students
wrote $x > 6$ or
 $x \leq 6$, lost
1 mark.

$\therefore x > 6$ is the largest positive
domain for an inverse, $\checkmark$

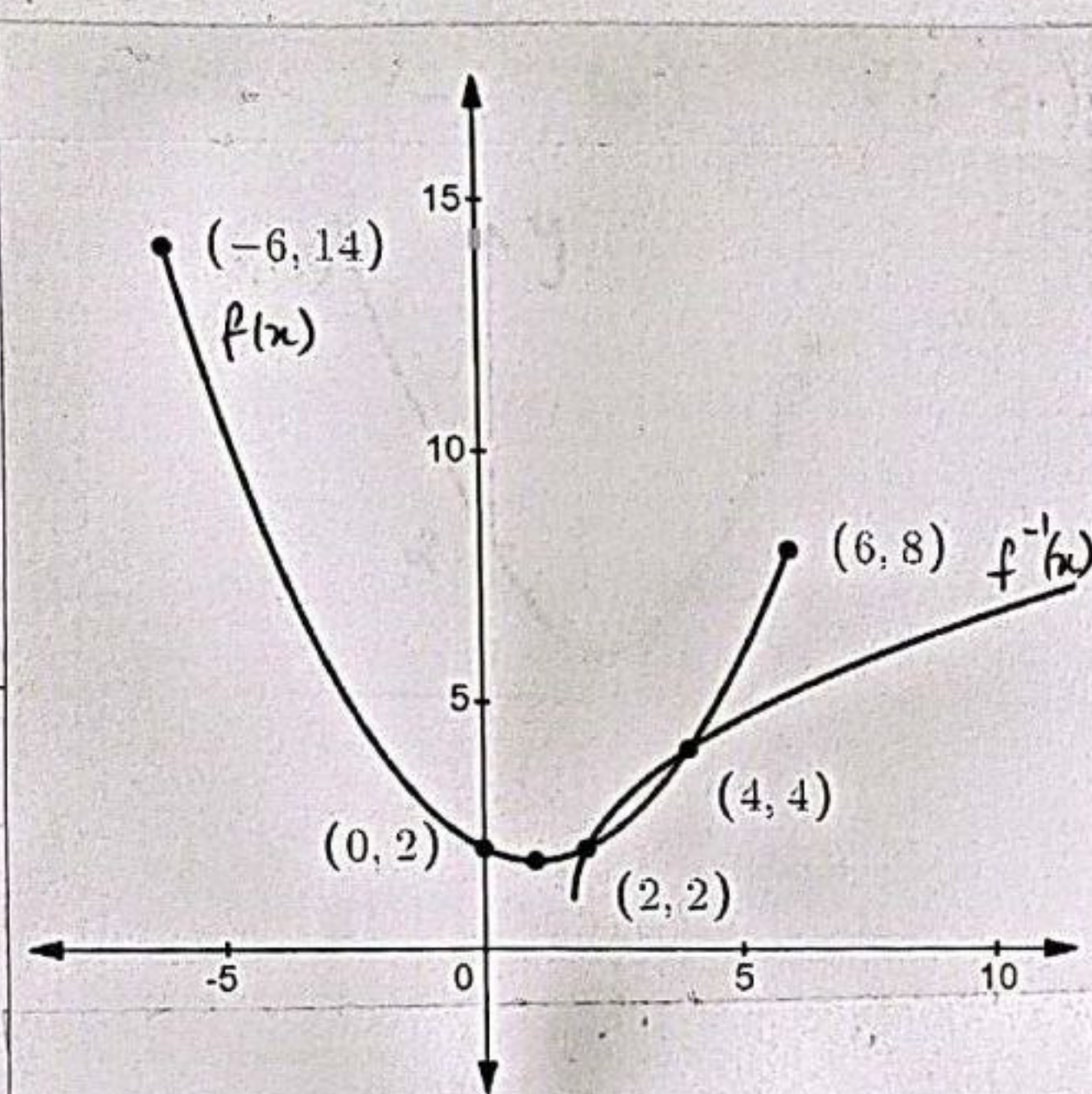
Question 14

a) $f(x) = \frac{1}{4}[(x-1)^2 + 7]$

i) y-intercept: $\frac{1}{4}[1^2 + 7] = 2$

vertex: $(1, \frac{7}{4})$

endpoints: $(-6, -14)$, $(6, 8)$



① - intercept + vertex labelled

① - endpoints labelled within domain

① - $[-6, 6]$

NOTE:

* Some students lost marks for messy diagrams.

The quality of some graphs is unacceptable.

ii) $f^{-1}(x): x = \frac{1}{4}[(y-1)^2 + 7]$

$$4x = (y-1)^2 + 7$$

$$4x - 7 = (y-1)^2$$

$$\pm \sqrt{4x-7} = y-1$$

$$y = 1 \pm \sqrt{4x-7} \quad \text{①}$$

since range is $[1, \infty)$

$$\therefore y = 1 + \sqrt{4x-7} \quad \text{①}$$

Note:

* Many students jumped

$$\text{to } y = 1 + \sqrt{4x-7}$$

without justifying why they chose that answer.

The ' $\pm$ ' step is needed.

* Some students wrote:

$$x = \frac{1}{4}[(y-1)^2 + 7]$$

$$\frac{x}{4} = (y-1)^2 + 7$$

iii) Points of intersection will occur at $y=x$ if they also exist the graphs $f(x)$ and $y=x$.

$$\textcircled{1} \quad y = \frac{1}{4} [(x-1)^2 + 7]$$

$$\textcircled{2} \quad y = x$$

equate $\textcircled{1}$ and $\textcircled{2}$:

$$x = \frac{1}{4} [(x-1)^2 + 7]$$

$$4x = (x-1)^2 + 7$$

$$4x = x^2 - 2x + 1 + 7$$

$$x^2 - 6x + 8 = 0$$

$$(x-4)(x-2) = 0$$

$$\therefore x = 4, 2$$

sub them into $\textcircled{2}$

$\textcircled{1} \therefore (4,4)$ and $(2,2)$ are the points of intersection for $f(x)$ and $y=x$,
thus by extension for $f(x)$ and $f^{-1}(x)$

NOTE:

* Many students made it difficult for themselves by trying to equate $f(x)$ and $f^{-1}(x)$.
Only a few were successful in getting the answer that way.

* Some students made the claim that there are no points of intersection without any working or justification
 $\therefore$ no CFE for their inverse graph.

v) on graph in i)

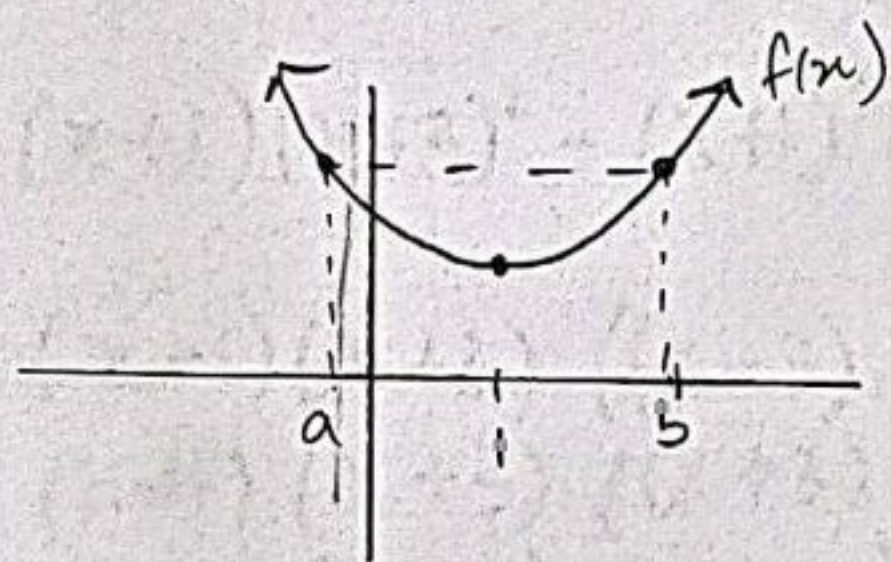
v)

Method ①:

Let $a < 1$

Now, from graph, there is a value b such that

$$f(a) = f(b)$$



And since $b > 1$

$$f^{-1}(f(b)) = b$$

$$\therefore f^{-1}(f(a)) = b$$

By symmetry, the midpoint between a and b is 1

$$\text{i.e. } \frac{a+b}{2} = 1 \quad \text{①}$$

$$b = 2 - a$$

$$\text{since } f^{-1}(f(a)) = b$$

$$\therefore f^{-1}(f(a)) = 2 - a \quad \text{①}$$

Method ②:

$$f(a) = \frac{1}{4} [(a-1)^2 + 7]$$

$$f^{-1}(f(a)) = 1 + \sqrt{4 \left(\frac{1}{4} (a-1)^2 + 7 \right) - 7}$$

$$= 1 + \sqrt{(a-1)^2 - 7 + 7}$$

$$= 1 + \sqrt{(a-1)^2} *$$

$$= 1 - (a-1)$$

$$= 2 - a \quad \text{// ①}$$

or if $= a$ ① only.

* Now:

$$\text{① } \sqrt{(a-1)^2} = a-1 \quad \text{if } a \geq 1$$

$$\text{or} \\ = -(a-1) \quad \text{if } a < 1$$

Since a is out of the domain, we take $a < 1$

$$\frac{1}{x+1} \leq \frac{x+1}{x+3} \quad \underline{x \neq -1, -3} \quad (1)$$

$$\frac{1(x+1)^2(x+3)^2}{(x+1)} \leq \frac{(x+1)(x+3)^2(x+1)^2}{(x+3)} \quad (1)$$

$$(x+1)(x+3)^2 \leq (x+1)^3(x+3)$$

$$(x+1)(x+3)^2 - (x+1)^3(x+3) \leq 0$$

$$(x+1)(x+3) [(x+3) - (x+1)^2] \leq 0$$

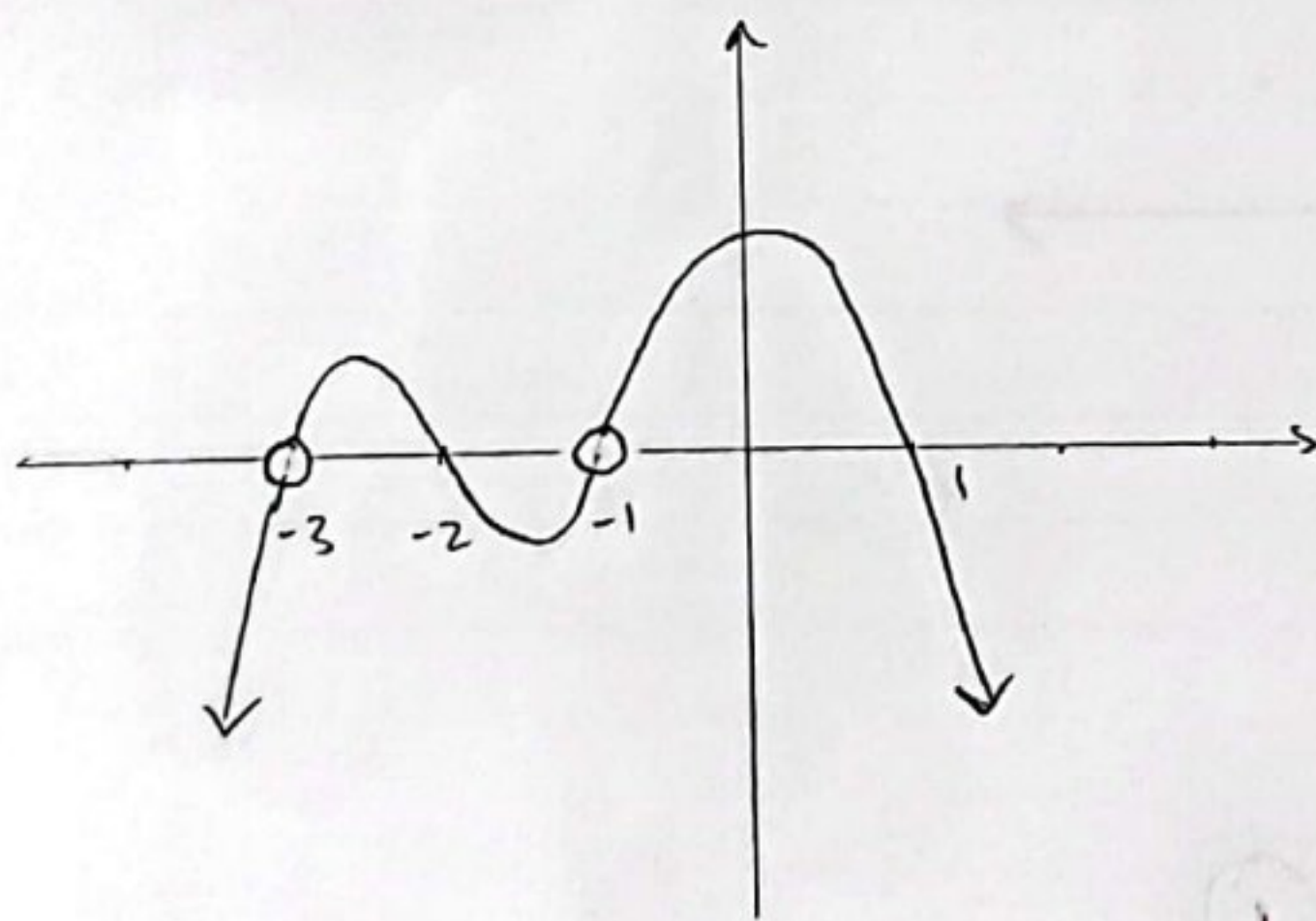
$$(x+1)(x+3) [(x+3) - (x^2+2x+1)] \leq 0$$

$$(x+1)(x+3) [(x+3) - x^2 - 2x - 1] \leq 0$$

$$(x+1)(x+3)(-x^2 - x + 2) \leq 0$$

$$-(x+1)(x+3)(x^2 + x - 2) \leq 0$$

$$-(x+1)(x+3)(x+2)(x-1) \leq 0 \quad (1)$$



$$x < -3, -2 \leq x < -1, x > 1$$

$$(-\infty, -3) \cup [-2, -1) \cup [1, \infty) \quad (1)$$

Note:

* Done badly due to poor algebra and understanding of inequalities.

* Students that eliminated solutions and made the inequality into a quadratic lost a mark for making it easier.

* Students that did the critical points method need to be careful to not skip a range of solutions.

e.g

